STAFF REVIEW · H446 144 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Internet layer.

2. Stack.

3. Tracking variable state changes as an algorithm runs.

4. Eight.

I Do — source worked steps

1. Name example.org — Identify the relevant information and make a prediction.

2. DNS returns an IP address — Explain the change using correct technical terms.

3. Connect to web server — DNS maps the hostname to relevant resource records such as an A/AAAA address. DNS does not directly transmit the web page. Browser makes a separate connection to the chosen server address and requests content, often over HTTPS.

We Do — guided CHECK

1. Cached old address
Identify the relevant information and make a prediction.
Server address changes
Explain the change using correct technical terms.
Refresh after TTL
The cached address may remain until its TTL expires; client may contact an outdated endpoint. The resolver can refresh after expiry. DNS propagation and authoritative changes do not guarantee immediate worldwide consistency.

You Do — source mark schemes

1. Hostname lookup [2], address records returned/cached [2], browser uses IP to connect to server [2].

2. DNS maps names/records [3], routers forward packets using IP destination/network information [3].

3. Resolver caching [2], TTL expiry [2], changed authoritative record [2], client/network diagnostic tests [2].

Exit Ticket — source answer

1. False. DNS supplies name-resolution data; web content is retrieved separately.